03 - Exploratory Data Analysis

Customer Churn Prediction | Repository-based interview study guide

Source: https://github.com/YuvarajGarla/Customer-churn-prediction

1. EDA goals

Exploratory data analysis is used to inspect data quality, understand feature distributions, identify unusual values, study relationships with the target, and develop hypotheses that can guide modelling. EDA findings should be treated as associations, not causal conclusions.

2. Initial inspection

```
telco_base_data.head()
telco_base_data.shape
telco_base_data.columns
telco_base_data.dtypes
telco_base_data.describe()
telco_base_data.info()
```

Operation	Purpose
head()	Inspect sample records
shape / columns	Understand dimensions and available attributes
dtypes / info()	Inspect types and non-null counts
describe()	Review numerical distributions and summary statistics

3. Target distribution

The notebook plots Churn counts and percentages. Its narrative identifies an approximately 73:27 majority-to-minority class ratio. This tells us to examine performance separately for churners rather than rely on accuracy alone.

4. Univariate and bivariate analysis

The EDA notebook uses count plots with hue=Churn to compare outcomes across categories such as Contract, PaymentMethod, and service subscriptions. It also uses density plots to compare MonthlyCharges and TotalCharges by churn outcome.

5. Main observations documented in the notebook

- Month-to-month contract customers show higher churn than longer-contract customers.
- Customers without OnlineSecurity and TechSupport show higher churn.
- Electronic-check users are identified as a high-churn group.
- Higher MonthlyCharges are associated with higher churn in the notebook analysis.
- Customers with shorter tenure / in the first year show higher churn.
- Longer-tenure customers show lower churn.
- Gender, PhoneService, and MultipleLines are described as having relatively weak association with churn.

Important: The observations above reflect the notebook's descriptive analysis. To compare churn rates fairly across categories, calculate within-category churn rates, not just raw counts. Category sizes differ, and

raw counts can be misleading.

6. TotalCharges and tenure relationship

The notebook notes that churn appears higher among customers with lower TotalCharges. This becomes easier to interpret alongside tenure: cumulative charges tend to increase with time subscribed and monthly price. A customer can therefore have high MonthlyCharges but low TotalCharges because their tenure is short.

TotalCharges is approximately related to MonthlyCharges × Tenure, but is not exactly equal because charges can vary over time and include other billing details.

7. Correlation analysis

```
telco_data_dummies.corr()['Churn'].sort_values(ascending=False)
sns.heatmap(telco_data_dummies.corr())
```

The notebook computes correlations after dummy encoding. For binary encoded variables, correlation can indicate association, but it is not causation. Correlation among one-hot columns and the target also needs context, including the reference categories and prevalence of each category.

8. Better EDA practices to mention

- Report churn rate within each category: P(Churn=1 | category), not only total counts.
- Use normalized stacked bars where category sizes differ.
- Inspect numeric distributions and outliers separately for churn/non-churn groups.
- Check association and redundancy among features.
- Do not use target information from outside the training data for any learned preprocessing decisions.

9. Interview-ready explanation

During EDA, I examined dataset structure, types, summary statistics, and the target distribution. I then compared churn and non-churn customers across contract, service, tenure, payment, and billing attributes. The notebook highlighted higher churn among month-to-month customers, customers without online security or technical support, electronic-check users, and customers with shorter tenure. These patterns helped frame the preprocessing and modelling work, while being treated as associations rather than causal effects.

10. Questions to prepare

Q. What is the difference between univariate and bivariate analysis?

Univariate analysis studies one variable's distribution. Bivariate analysis studies the relationship between two variables, for example MonthlyCharges and Churn.

Q. What does a KDE plot show?

It estimates a smoothed probability density function for a continuous variable, allowing comparison of distributions across groups.

Q. Why can raw category counts be misleading?

Categories have different population sizes. A larger category can produce more churners even if its churn rate is lower. Compare normalized rates.

Q. Does correlation imply causation?

No. A correlation can arise from confounding, selection, reverse relationships, or coincidence. Causal claims need an appropriate causal design.